

RC-NFGD | reliability-gated force feedback

1,000-step reverse diffusion

window enlarged for clarity

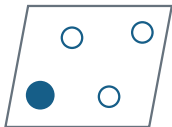
last 20 predictors only

$t = 1$

$t = 0.02$

ϵ

01 CLEAN STRUCTURE



predicted clean crystal
from current MatterGen state

02 FORCE FEEDBACK



frozen MatterSim-5M
center forces $\rightarrow$ bounded δr

03 SCORE UPDATE

Cartesian $\rightarrow$ fractional

$$\delta \mathbf{f} = \delta \mathbf{r} \cdot \text{inv}(\mathbf{H}_0)$$

position predictor only

$$\tilde{\mathbf{s}}_t^f = \mathbf{s}_t^f + \delta \mathbf{f} / \mathbf{c}_t^f$$

atomic / cell / corrector unchanged



ACCEPT finite output, valid cell / score coefficient, and distance $\geq 0.5 \text{ \AA}$

Otherwise keep the original MatterGen score; no feedback is applied.

$$\eta = 0.005 \text{ \AA} \cdot \text{hard cap} = 0.01 \text{ \AA} \cdot \text{CFG} = 2.0$$